

Module 5: Inverses and Radicals – Prequiz

Name:

Date:

Four short questions. Remember that an inverse function reverses the original input-output rule.

1. Find $f^{-1}(x)$ for $f(x) = 4x + 1$.
2. If $f(x) = 2x - 3$, find $f^{-1}(9)$.
3. Explain why $g(x) = x^2 + 4$ does not have an inverse function on all real numbers. If the domain is restricted to $x \geq 0$, find $g^{-1}(x)$ and state its domain.
4. Solve $\sqrt{3x + 1} = 7$ and check your solution.

Module 5: Inverses and Radicals – Weekly Check-In

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Three questions. Remember that an inverse reverses the original input-output rule.

1. Finding and using an inverse.

Let

$$f(x) = 4x - 7.$$

(a) Find $f^{-1}(x)$.

(b) Find $f^{-1}(13)$.

(c) Verify your inverse by simplifying $f(f^{-1}(x))$.

2. Restricting a domain.

Explain why $g(x) = x^2 + 2$ does not have an inverse function on all real numbers. Restrict the domain to $x \geq 0$, find $g^{-1}(x)$, and state the domain of the inverse.

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3. Solving a radical equation.

Solve

$$\sqrt{2x+3} = x.$$

Show the algebra that produces all candidates, then check each candidate in the *original* equation and identify any extraneous solution.